Compressing the Latent Communication Channel of a Recursive Multi-Agent System: An Empirical Study

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Abstract

We ask whether the latent vectors that agents in a recursive multi-agent system exchange can be compressed without harming downstream accuracy. The question is not obvious: unlike the KV cache — the usual target of activation quantization, where each entry is averaged over many attention queries — a message in this latent channel is consumed once, without averaging, through a bottleneck, and repeatedly along the recursion. We inject the MSE-optimal core of TurboQuant [Zandieh et al., 2026] — a data-oblivious Haar rotation followed by per-coordinate Lloyd–Max scalar quantization, without the QJL inner-product residual — into every latent link of RecursiveMAS Sequential-Light [Yang et al., 2026], and measure its effect on math500 at three granularities: per-link reconstruction error, end-to-end accuracy, and the per-step output distribution. The per-link error tracks the quantizer's reported distortion rate ($E_{\mathrm{rel}} \approx 0.009$ at 4 bits), and under sampled decoding we detect no accuracy change from 4× to 16× compression at n=250. Our most informative result is a qualifier: under deterministic greedy decoding the compressed channel is *answer-preserving but not trajectory-preserving* — the 4-bit channel changes the generated reasoning text in 88% of problems while preserving the final answer, and a strict ±2-point paired equivalence test is inconclusive ($\Delta = -2.0$ pp, 95% CI $[-6.4, +2.4]$). This is a focused study of one released system on one benchmark under fake-quantization; we report what we measured and bound what we did not.

1 Introduction

Multi-agent systems built from heterogeneous language models increasingly communicate in latent space rather than in text: one agent projects its last-layer hidden state and hands it to another. RecursiveMAS [Yang et al., 2026] formalizes this with a lightweight recursive link that transmits an agent's hidden state across agents of different hidden sizes. As such systems move toward distributed deployment, the bandwidth of this latent channel becomes a practical cost, which raises a concrete question: how compressible is it?

Activation quantization is well studied for the KV cache, which is unusually tolerant. Each cached key/value is averaged over many attention queries, softmax attenuates per-token noise, and the outliers are structured. The latent communication channel has none of these properties. It is a one-shot message — the receiving agent consumes it once, with no averaging. It is an information-dense bottleneck (a two-layer residual MLP with LayerNorms). And it is traversed repeatedly along the recursion, so per-call errors could in principle compound. A priori it is unclear whether this channel behaves like the compressible KV cache or like a fragile final-layer logit.

We answer the question empirically. We take TurboQuant's MSE-optimal quantizer unchanged from Zandieh et al. [2026] — a data-oblivious Haar rotation, which concentrates the per-coordinate marginal, followed by an optimal scalar quantizer — and inject it into every link of RecursiveMAS Sequential-Light. We then measure math500 accuracy and output fidelity, and we contrast sampled with deterministic greedy decoding so that we can attribute effects cleanly.

Contributions.

- We frame the latent communication channel of a recursive multi-agent system as a compression target distinct from the KV cache, and give an instrumentation-only methodology: the model and the quantizer are untouched.
- Under sampled decoding, we detect no math500 accuracy change from 4× to 16× compression at n=250 (two-proportion z-tests, all $p > 0.4$).
- Under deterministic greedy decoding, the compressed channel is *answer-preserving but not trajectory-preserving*: the 4-bit channel changes the generated text in 88% of problems, yet the paired accuracy difference is small and a ±2-point equivalence test is inconclusive ($\Delta = -2.0$ pp, CI $[-6.4, +2.4]$). A REF-vs-REF control attributes the change to the quantizer rather than to nondeterminism.
- We document two reproducibility hazards: bfloat16 silently collapses this recursive pipeline on pre-Ampere GPUs, and an injected quantizer's internal dtype must match the pipeline's.

This is a study, not a new method: the quantizer and the system are both taken unchanged from prior work [Zandieh et al., 2026, Yang et al., 2026]. We evaluate the MSE core of TurboQuant, not the complete algorithm with its QJL residual (Section 3).

2 Background and related work

Activation and KV-cache quantization. A substantial literature compresses LLM key/value caches to low bit-rates with little quality loss, exploiting the statistical redundancy and averaging inherent to attention. Our target differs structurally — a one-shot, bottlenecked, repeatedly-traversed latent message rather than an averaged cache entry — so tolerance does not transfer by assumption, which is what motivates a direct measurement.

Data-oblivious vector quantization. TurboQuant [Zandieh et al., 2026] attains near-optimal distortion with a random rotation followed by an optimal per-coordinate scalar quantizer, and requires no calibration data. It has two components: an MSE-optimal quantizer, and a QJL inner-product residual for inner-product-preserving tasks. We use only the MSE-optimal core (Section 3); the data-oblivious property is what makes a drop-in deployment plausible in the first place.

Recursive multi-agent systems. RecursiveMAS [Yang et al., 2026] connects heterogeneous agents through a recursive latent link. We treat its released Sequential-Light constellation as a fixed system and study only the compressibility of its latent communication channel; we modify neither the agents nor the link weights.

3 Method

The quantizer. We instantiate the MSE-oriented core of TurboQuant, which we refer to as TurboQuant-MSE. Given a channel vector $x \in \mathbb{R}^d$, TurboQuant-MSE normalizes x onto the unit sphere, applies a fixed Haar-random rotation R (computed once and stored as a buffer), quantizes each rotated coordinate with a 2^b-level Lloyd–Max codebook precomputed for the concentrated marginal, then inverts the rotation and rescales. The quantizer is data-oblivious, carries no per-tensor scale or group size, and works at any d. We do *not* include TurboQuant's QJL inner-product residual, which targets inner-product preservation; our intervention replaces dense latent vectors at the communication boundary, so it targets direct reconstruction fidelity. This narrows the claim to a TurboQuant-MSE instantiation rather than the complete TurboQuant algorithm, and it sharpens the empirical point: the MSE core alone suffices for the final-answer robustness we measure. We verified our implementation against the reference numerically (codebook and rotation match to 10^{-4}, end-to-end reconstruction to 5×10^{-4}) and against the reported distortion rate (Section 4.1).

Insertion point. RecursiveMAS routes each latent transfer through an adapter that ends in a LayerNorm. We wrap each adapter's forward pass so its post-LayerNorm output passes through TurboQuant-MSE before being returned. Quantization runs in fp32, and the result is cast back to the pipeline dtype. The wrapper is reversible and records per-call distortion; the model and quantizer code are never edited.

Measurement. We evaluate at three granularities. (i) *Per-link distortion*: the relative squared reconstruction error $E_{\text{rel}} = \|x - \hat{x}\|_2^2 / \|x\|_2^2$ and the cosine similarity, at every adapter call. (ii) *End-to-end accuracy* on math500. (iii) *Per-step output fidelity*: for a paired unquantized (REF) and 4-bit (INT4) run under greedy decoding with a shared seed, we capture the top-K=512 next-token logits at each decode position and compute the KL divergence, the Jensen–Shannon divergence, and a probability-space MSE between the two distributions. Because the two greedy trajectories can diverge — after which a positional comparison is meaningless — we report (iii) only over the matched prefix (the positions up to and including the first token at which the two runs choose different tokens), and we report the divergence rate separately. We sweep the channel-traversal count $T = \texttt{num\_recursive\_rounds} \in \{1, 2, 3, 4\}$, and we compare accuracy with a paired bootstrap and a two one-sided test (TOST) for equivalence at a *pre-specified* margin of ε=2 percentage points (pp).

A note on the equivalence verdict. TOST certifies equivalence only when both one-sided tests reject. When neither rejects, equivalence is simply *not established*; we report this as "inconclusive" and never as evidence that the two conditions differ.

4 Experiments

All runs use math500, seed 42, and fp32. Compute is a single A100-40GB (Modal) for the paired greedy analysis and a single T4 (Kaggle, fp32) for the sampled ladder. Two decoding regimes carry different baselines, and we never compare across them (Table 1): the sampled ladder uses the released sampling settings with *unpaired* runs, while the fidelity and equivalence analyses use greedy decoding with *paired* REF/INT4 runs on identical inputs. A separate n=50 exploration baseline of ∼84% reflects an easier subset and is not comparable to either powered baseline.

4.1 Per-link distortion tracks the reported rate

On the real Solver adapter output (dimension 1536), the relative squared error of TurboQuant-MSE follows the same rate curve as a synthetic Gaussian baseline and the values reported by Zandieh

Table 1: Baselines are experiment-specific; we do not compare across decoding regimes. Sampled = unpaired runs under the released sampling settings; greedy = paired REF/INT4 on identical inputs.

experiment	decoding	n	baseline	treatment
sampled ladder (T4 fp32)	sampled	250	75.2%	8/4/2-bit: 78.4/76.8/75.2%
greedy paired (A100 fp32, T=3)	greedy	250	REF 76.8%	INT4 74.8%
REF-vs-REF control	greedy	250	76.8%	76.8% (Δ0)
depth sweep	greedy	50	per-T	INT4

Table 2: Per-link relative squared reconstruction error $E_{\text{rel}} = \|x - \hat{x}\|_2^2 / \|x\|_2^2$ of TurboQuant-MSE on the real adapter output, versus synthetic data and the reported TurboQuant distortion rate (lower is better).

bits	synthetic (d=2048)	real adapter (d=1536)	reported [Zandieh et al., 2026]
8	0.0001	0.0001	(≈lossless)
4	0.0095	0.0093	0.009
3	0.0345	0.0339	0.030
2	0.1175	0.1159	0.117

et al. [2026]; agreement is close at the bit-rates used downstream (Table 2), tightest at 8 and 4 bits and looser at 3 bits. The latent channel is therefore compressible by TurboQuant, and the data-oblivious property holds on a real channel without calibration.

4.2 Sampled decoding: no detected accuracy change at $4\times$–$16\times$

We inject TurboQuant-MSE at every link and evaluate math500 at n=250 under sampled decoding. We do not detect an accuracy change from the unquantized baseline at any bit-rate from 8 down to 2 (Table 3 and Figure 1; two-proportion z-tests, all $p > 0.4$; the 95% Wald intervals for every delta straddle zero). At 2 bits per coordinate — a $16\times$ reduction, near the information-theoretic floor — the quantized system reaches the same accuracy count as baseline (188/250). This is consistent with no degradation at the resolution of this experiment; it is not a proof of equality (Section 4.3 probes the channel more finely).

What this buys, information-theoretically. For the powered T=3 configuration, the all-link run quantizes 1,152 latent token-vectors per math500 problem (288,000 over 250 problems). With a representative channel dimension $d \approx 2048$ (agent hidden sizes span 1536–2048), the per-problem latent traffic is

$$\text{traffic} \;=\; N_{\text{vectors}} \times d \times \tfrac{b}{8} \;\approx\; 1{,}152 \times 2048 \times \tfrac{b}{8} \text{ bytes},$$

i.e. ≈ 9.0 MiB at fp32 (b=32), ≈ 1.1 MiB at 4 bits, and ≈ 0.56 MiB at 2 bits. These are information-theoretic, packed-integer figures; we measure fake-quantization fidelity, not wall-clock bandwidth or latency (Section 6).

4.3 Greedy decoding: answer-preserving, not trajectory-preserving

Accuracy is a one-bit-per-problem metric. To probe the channel more finely we run a paired REF-vs-INT4 comparison under deterministic greedy decoding at n=250, T=3. The 4-bit run scores

Table 3: Math500 accuracy at n=250 under sampled decoding (T4, fp32). We detect no difference from baseline at any bit-rate (two-proportion z-test, all $p > 0.4$); 95% intervals are Wald two-proportion-difference intervals.

bits/coord	compression	accuracy	Δ baseline (95% CI)	p
– (baseline)	1×	75.2%	–	–
8	4×	78.4%	+3.2 [−4.2, +10.6]	> 0.4
4	8×	76.8%	+1.6 [−5.9, +9.1]	> 0.5
2	16×	75.2%	0.0 [−7.6, +7.6]	1.0

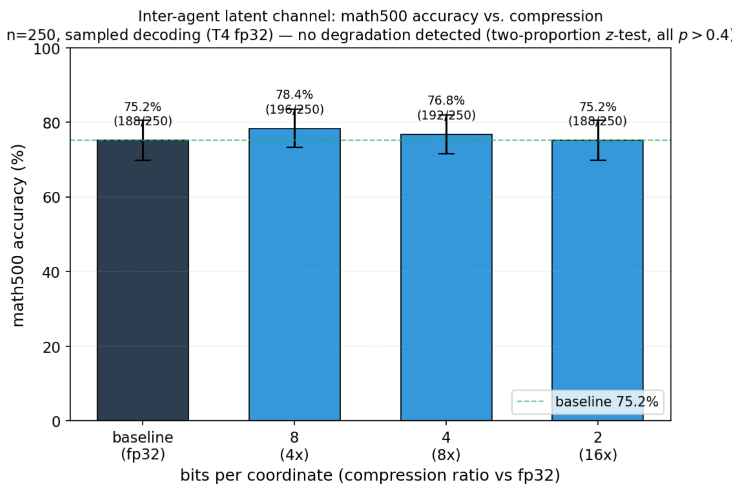

Figure 1: Math500 accuracy versus TurboQuant-MSE bit-rate at n=250 under sampled decoding. The curve is flat from the unquantized baseline through 16× compression; every bit-rate lies within sampling noise of the baseline. Under greedy decoding a stricter paired equivalence test is inconclusive (Section 4.3).

74.8% against the unquantized 76.8%: a paired difference of −2.0 pp with a 95% bootstrap interval of $[-6.4, +2.4]$. At the pre-specified ε=2-pp margin the equivalence test is inconclusive — the estimate sits on the margin and the interval reaches −6.4 pp, so we establish neither equivalence nor a difference (Table 4). The sampled ladder of Section 4.2 and this greedy estimate are not directly comparable (different decoding, different unpaired vs paired runs); both effects are small, and we summarize across regimes as: the 4-bit accuracy effect is at most a few points and not robustly distinguishable from zero.

The effect is the quantizer, not nondeterminism. We run the unquantized configuration twice. The two runs are bit-identical: zero problems differ, zero trajectory divergence, zero per-step KL (Table 4, control row). The pipeline is therefore deterministic across containers, and any REF-vs-INT4 difference is attributable to the quantizer rather than to run-to-run noise.

The quantizer changes the trajectory but not the answer. Although the accuracy effect is small, the 4-bit channel changes the generated tokens substantially. On the clean n=250 run, the greedy trajectory eventually selects a different token from the unquantized run in 88% of problems (divergence rate 0.88), after a mean matched prefix of ≈ 80 tokens (against a control divergence

Table 4: Paired accuracy and TOST equivalence (ε=2pp) under greedy decoding at T=3, n=250. "Inconclusive" means equivalence was not established, not that the conditions differ.

comparison	Δacc (95% CI)	TOST verdict	n
INT4 vs REF	-2.0 $[-6.4, +2.4]$	inconclusive	250
REF vs REF (control)	0.0 $[0.0, 0.0]$	—	250

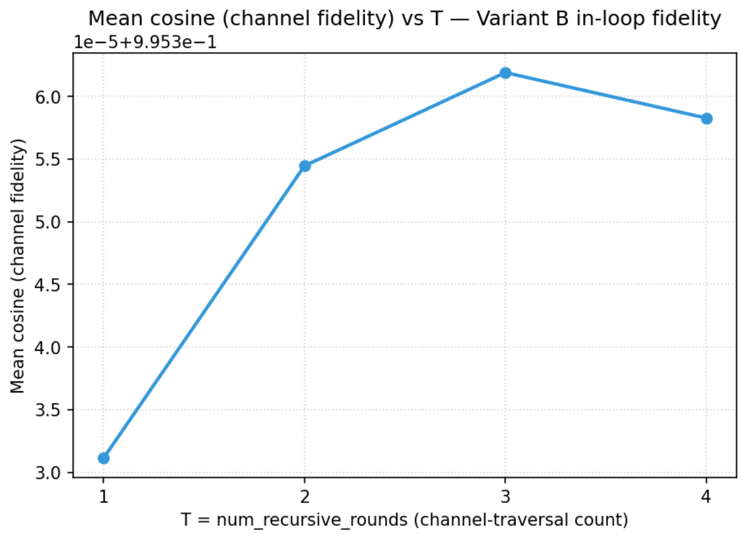

Figure 2: Per-call channel cosine (4-bit round-trip) versus recursion depth T. The cosine is ≈ 0.9954 at every T — a consistency check, since the data-oblivious quantizer's per-call distortion is depth-independent by construction, not a depth result.

rate of zero). Over those matched prefixes the output distributions are measurably but modestly shifted, not zero: matched-prefix KL 0.325 nats (95% CI $[0.162, 0.512]$) and Jensen–Shannon 0.032 ($[0.018, 0.047]$), with a per-call channel cosine of 0.9954 (relative L2 ≈ 0.096, consistent with the 4-bit rate). The 4-bit channel thus changes the *reasoning text* in most problems while the *final answer* survives: the compressed channel is answer-preserving, not trajectory-preserving.

4.4 Depth and link localization

Depth. Across $T \in \{1, 2, 3, 4\}$ the system's accuracy does not collapse, so the catastrophic per-call-error accumulation one might fear a priori ($\sim \varepsilon \cdot N$ over N channel calls) does not occur. We avoid a stronger depth claim. The per-call channel cosine is 0.9954 at every T (Figure 2), but this is depth-independent *by construction* — the quantizer is data-oblivious, so its per-call distortion depends only on bit-rate and dimension — and is a consistency check, not a depth result. Our finer per-step KL estimator is too unstable to resolve a depth trend: at the same T=3 it reads 0.03 at n=50 and 0.33 at n=250, and it averages over run-dependent positions.

Localization. The channel splits into same-model inner links (intra-agent recursion, $\sim$12,544 calls) and cross-model outer links (the genuinely inter-agent transfers, $\sim$256 calls). We quantize each in isolation to ask which carries the small accuracy cost (Table 5). The question is not resolved at n=250: the three paired deltas (-0.8, -1.2, -2.0 pp) all have 95% intervals that straddle zero

Table 5: Selective quantization at T=3, n=250, greedy (clean retry). The paired accuracy deltas do not resolve whether inner or outer links carry the small cost: all 95% intervals straddle zero and overlap.

quantized	#calls	accuracy	Δacc vs REF (95% CI)
REF (none)	0	76.8%	–
inner only	12,544	76.0%	−0.8 [−5.2, +3.6]
outer only	256	75.6%	−1.2 [−5.6, +3.2]
both	12,800	74.8%	−2.0 [−6.4, +2.4]

and overlap heavily, so we cannot attribute the cost to either link type. The divergence-rate metric does not help either — it is a saturating binary near 0.88 for the all-link run and cannot separate conditions. Both link types plausibly contribute; resolving their relative contribution would require a position-aligned (teacher-forced) per-step measurement that we did not run.

4.5 Reproducibility hazards

bfloat16 on pre-Ampere GPUs. The released checkpoints specify bfloat16, and the default loader uses it. On GPUs without native bf16 support (compute capability below 8.0: P100, V100, T4) the recursive rollouts silently collapse math500 accuracy to ∼30–35%; forcing fp32 restores it to ∼84% on the same hardware. The collapse is a deep-recursion effect; single-forward use is unaffected.

Dtype mismatch. Placing the fp32-internal quantizer inside a bf16 pipeline introduces a cast between bf16 and fp32 at each of ∼48 calls; the casts accumulate $\sim 10^{-3}$ error per call and manufacture a spurious −19-point accuracy drop. A dtype-coherent (fp32) pipeline removes it. An injected quantizer's internal dtype must match the pipeline it is measured in.

5 Discussion

What the result means. The compressed channel is answer-preserving but not trajectory-preserving. Aggressive data-oblivious quantization of the latent messages does not keep the generated reasoning identical — it changes the trajectory in most problems — yet the final answer survives. The robustness we measure is therefore a property of the *task and system*, whose redundancy lets a perturbed latent thought still land on the right answer, rather than evidence of a noise-free channel.

Does the error propagate? Yes, and we do not claim otherwise. The 4-bit perturbation propagates far enough to change the greedy trajectory in 88% of problems and to shift the matched-prefix output distribution (KL ≈ 0.33 nats). What we observe is that this propagation does not *amplify* into an accuracy collapse as recursion depth grows to T=4, and that per-call input distortion is bounded and depth-independent by construction. We cannot certify that the propagated distortion is uniformly small: our per-step KL estimate is unstable across sample sizes (0.03 at n=50 vs 0.33 at n=250), and we did not run a teacher-forced per-step measurement.

Why it matters. Latent communication is proposed as the efficient alternative to text-based agent messaging. If that latent channel is *also* highly compressible at no detected accuracy cost, the efficiency argument compounds. This study gives narrowly-scoped evidence that a real recursive multi-agent latent channel tolerates 4×–16× data-oblivious compression, and it characterizes the failure mode — trajectory drift under preserved answers — that a deployment would need to monitor. It does not establish a deployable bandwidth saving, generality beyond this system, or unconditional losslessness.

6 Limitations

We measure *fake*-quantization fidelity: the dequantized vector is exactly what a packed low-bit transport would deliver, so the accuracy and distribution numbers are faithful, but we report no measured wall-clock bandwidth, latency, or memory saving; the traffic figures in Section 4.2 are information-theoretic. We evaluate only the MSE core of TurboQuant; the QJL inner-product residual is an untested future ablation, especially at lower bit-rates or inner-product-dominated objectives. The accuracy equivalence analysis is at n=250 for T=3; the depth sweep is at n=50, which underpowers a per-T equivalence test. Our per-step KL is a matched-prefix proxy rather than a position-aligned (teacher-forced) measurement, and is too unstable to support a depth or localization conclusion. All results use a single seed, a single benchmark (math500), a single subset ordering, and a single released constellation (Sequential-Light); we make no claim of generality. Finally, the quantizer and the system are both taken unchanged from prior work; the contribution is the measurement, not a method.

7 Conclusion

The latent communication channel of a recursive multi-agent system — one-shot, bottlenecked, and repeatedly traversed — is highly compressible by the MSE core of a data-oblivious quantizer: under sampled decoding we detect no math500 accuracy change from 4× to 16× at n=250, and accuracy does not collapse as recursion depth grows. The honest qualifier is that under deterministic greedy decoding the compressed channel is answer-preserving but not trajectory-preserving: it changes the generated reasoning text in most problems and yields a small, non-significant accuracy difference whose equivalence we cannot certify. The main practical obstacle we met was not the quantizer but dtype handling, which we document. Code, the reproducibility workflow, and post-hoc analysis scripts for every experiment accompany this write-up.

References

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